

Four-Wave Mixing in Hot ^{85}Rb : A Physics-First Walkthrough of the GABES Engine

Current FWM Physics Model

Personal study notes

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Abstract

These notes derive, step by step, every piece of physics that the GABES four-wave-mixing (FWM) code evaluates — from the open-system master equation of the ^{85}Rb D_1 double- Λ system, through the Doppler average and the truncated-Floquet sideband steady state, to the Maxwell–Bloch propagation that turns the atomic susceptibility into seed/conjugate gain, intensity-difference squeezing, and (in the second mode) a spontaneous-FWM biphoton waveform. Seeded squeezing includes the reference 0.3° pump–probe longitudinal mismatch in the app’s fast and balanced paths; Fine mode adds a conservative χ -reused refractive correction and segmented propagation. Biphoton mode uses vector phase matching: calibrated longitudinal Δk_z plus absolute transverse Δk_x , with a two-dimensional signal–idler vector acceptance map in Fine mode. The heavy OBE grid can be evaluated by optional compiled Numba kernels, while the (Δ, T) squeezing scan reuses one master $\bar{\chi}(\delta, \Delta_{\text{eff}})$ table. The emphasis is on the *physics*; the code is referenced only as a map so that each equation can be located in `gabes/`. The direct `compute_spectrum` path is regression-anchored to G. Sim, H. Kim, and H. S. Moon, *Sci. Rep.* **15**, 7727 (2025).

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1 Purpose and the object we compute

Non-degenerate four-wave mixing in a hot alkali vapour is a third-order parametric process. A strong *pump* drives a multilevel atom; a weak *seed* (probe) injected on one Raman leg stimulates the emission of a *conjugate* on the other leg. Energy and momentum conservation tie the two generated photons together, so signal and conjugate emerge as a two-mode squeezed (twin-beam) state: their *intensity difference* is quieter than the shot-noise limit, even though each beam alone is bright and thermal-like.

The code answers two questions:

1. **Seeded / squeezing mode.** Given a one-photon detuning Δ , cell temperature T , pump power, seed power, and detection efficiency, what is the seed power gain G_s , the conjugate gain G_c , and the intensity-difference squeezing S (in dB) as a function of the two-photon detuning δ ? This is the physics of Sim *et al.*, and is the regression-anchored core.
2. **Spontaneous / biphoton mode.** For a cascade or diamond topology with *no* seed, what is the time-correlation $g_{SI}^{(2)}(\tau)$ of the spontaneously generated signal-idler pair, its width, pair rate, and coincidence-to-accidental ratio (CAR)? This is a reference-calibrated source estimate, not a full quantum-Langevin propagation.

The whole numerical machine exists to evaluate one object accurately: the *atomic susceptibility matrix* $\bar{\chi}(\delta, \Delta; v, T)$ that links the induced optical polarizations on the two Raman legs to the seed and conjugate fields. Everything before it (Hamiltonian, dissipator, Floquet, Doppler) builds that matrix; everything after it (Maxwell-Bloch, saturation, squeezing formula) reads it out.

Code map. The seeded pipeline lives in `gabes/schemes/fwm.py::compute_spectrum`; the physics-agnostic OBE engine is `gabes/core.py`; the susceptibility $\rightarrow$ observable layer is `gabes/observables.py`; the level model and rates are `gabes/atoms.py` and `gabes/hyperfine.py`; the thermal average is `gabes/doppler.py`; physical constants and field helpers are `gabes/constants.py`. The biphoton mode is the `GenericFWMSolver` class in the same `fwm.py`.

2 The double- Λ system and its energy scales

2.1 Levels

The seeded model uses the ^{85}Rb D_1 line ($5S_{1/2} \rightarrow 5P_{1/2}$, $\lambda \approx 795\text{ nm}$) reduced to four levels (`atoms._double_lambda_rb85`):

index	state	manifold	role
g_1	$ 5S_{1/2}, F=2\rangle$	ground	lower Raman state
g_2	$ 5S_{1/2}, F=3\rangle$	ground	upper Raman state
e_2	$ 5P_{1/2}, F'=2\rangle$	excited	optical
e_3	$ 5P_{1/2}, F'=3\rangle$	excited	optical

The two ground hyperfine states are split by $\nu_{\text{HF}} = \nu_{\text{ground HF}} = 3.0357\text{ GHz}$; the two excited states by $\nu_{\text{HF}}^{(e)} = 361.58\text{ MHz}$. The geometry is “double- Λ ”: each ground state couples optically to

both excited states, and the two ground states share the excited manifold. That shared manifold is what lets a seed on the $g_2 \leftrightarrow e$ leg transfer coherence to the $g_1 \leftrightarrow e$ leg and generate the conjugate — the four “waves” are pump, pump, seed, conjugate.

2.2 Hyperfine line strengths (Clebsch–Gordan)

Each $F \rightarrow F'$ link carries a relative strength C_F^2 from the Wigner $3j/6j$ sum (`hyperfine.CF2`), exact rationals for ^{85}Rb D_1 :

$$C_F^2(2, 2) = \frac{10}{81}, \quad C_F^2(2, 3) = \frac{35}{81}, \quad C_F^2(3, 2) = \frac{35}{81}, \quad C_F^2(3, 3) = \frac{28}{81}. \quad (1)$$

These factors enter the physics in three distinct places, and using one consistent quantity for all three is what makes the model self-consistent:

1. **Coherent drive.** The effective Rabi frequency on link $g \rightarrow e$ is scaled by $\sqrt{3 C_F^2(F_g, F_e)}$ (`TRANSITION_DIPOLE_SCALE`); the factor 3 is the $J \rightarrow J'$ reduction of the D_1 reduced matrix element.
2. **Polarization readout.** The macroscopic polarization on a leg is the same CG-weighted sum of optical coherences (Sec. 7).
3. **Spontaneous emission branching.** An excited state e decays to ground g at the branched rate

$$\Gamma_{e \rightarrow g} = \Gamma \frac{C_F^2(F_g, F_e)}{\sum_{g'} C_F^2(F_{g'}, F_e)}, \quad \sum_g \Gamma_{e \rightarrow g} = \Gamma, \quad (2)$$

so total population leaves each excited state at the natural rate Γ (`atoms._double_lambda_rb85`).

2.3 Energy scales and why the rotating frame is mandatory

The relevant frequencies span fifteen orders of magnitude: the optical carrier is $\omega_{D_1}/2\pi = 377$ THz, the hyperfine splitting 3 GHz, the natural linewidth $\Gamma/2\pi = 5.746$ MHz, the ground-coherence decay 100 kHz. No solver integrates the bare optical oscillation. Instead we move to a frame co-rotating with the pump and apply the rotating-wave approximation (RWA): all optical carriers are removed, leaving *detunings* of order MHz–GHz on the diagonal and slowly varying Rabi couplings off-diagonal. Throughout, energies are expressed as angular frequencies (i.e. $\hbar \equiv 1$ in the Hamiltonian), so “ H ” below is really $H/\hbar$ in rad s^{-1} .

3 Units, fields, and the vapour density

3.1 Rabi frequency from beam power

A beam of power P and $1/e^2$ intensity radius w_0 has peak intensity $I = 2P/(\pi w_0^2)$. The code converts this to a Rabi frequency through the saturation intensity (`constants.rabi_freq`):

$$I = \frac{2P}{\pi w_0^2}, \quad \Omega = \Gamma \sqrt{\frac{I}{2I_{\text{sat}}}}, \quad I_{\text{sat}} = 4.484 \text{ mW cm}^{-2}. \quad (3)$$

Equation (3) is just the standard two-level relation $I/I_{\text{sat}} = 2\Omega^2/\Gamma^2$ solved for Ω ; tying Ω to I_{sat} keeps the drive strength anchored to a measured quantity rather than to an absolute dipole. With the paper geometry ($w_{\text{pump}} = 530 \mu\text{m}$, $w_{\text{seed}} = 330 \mu\text{m}$, both $1/e^2$ radii), 600 mW of pump gives $\Omega_{\text{pump}}/2\pi$ of order a GHz, while 8 μW of seed gives $\Omega_{\text{seed}}/2\pi$ of order MHz — a clean separation of “strong pump, weak seed” that the linear-response treatment below relies on.

3.2 Number density

The atomic density sets the optical depth and hence the gain. For the seeded model the Steck liquid-phase vapour-pressure fit is used at natural abundance (`atoms.rb85_density`):

$$\log_{10} P_{\text{torr}} = 15.88253 - \frac{4529.635}{T} + 0.00058663 T - 2.99138 \log_{10} T, \quad (4)$$

$$N(^{85}\text{Rb}) = f_{85} \frac{P}{k_B T}, \quad f_{85} = 0.7217, \quad P = P_{\text{torr}} \times 133.322 \text{ Pa}. \quad (5)$$

The exponential temperature dependence of P is the origin of the model’s extreme sensitivity to T : a few degrees changes N by tens of percent, and the gain is exponential in N (Sec. 9). The biphoton mode uses the elemental densities in `species.number_density` (CRC fit for Rb, Steck for Cs), with the per-isotope share applied separately.

4 Open-system dynamics: the master equation

The atom is open: it spontaneously emits, and ground coherences dephase from collisions and transit. The state is therefore a density matrix ρ obeying a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation,

$$\dot{\rho} = -i[H, \rho] + \mathcal{D}[\rho], \quad (6)$$

with H the rotating-frame Hamiltonian (Sec. 4.1) and $\mathcal{D}$ the dissipator (Sec. 4.2). We only ever want the *steady state* ($\dot{\rho} = 0$), since the experiment is continuous-wave.

4.1 The Hamiltonian

In the rotating frame the diagonal of H carries detunings and hyperfine splittings, and the off-diagonal carries the (RWA) drives. The static part at zero effective detuning is built by `static_hamiltonian_at_Deff_zero`:

$$H_0 = \underbrace{\delta |g_2\rangle\langle g_2| - \omega_{\text{HF}}^{(e)} |e_2\rangle\langle e_2|}_{\text{diagonal: two-photon \& excited HF}} + \underbrace{\sum_{\text{legs}} \frac{\Omega_{ge}}{2} (|g\rangle\langle e| + |e\rangle\langle g|)}_{\text{RWA drives}}, \quad (7)$$

where δ is the two-photon (Raman) detuning, $\omega_{\text{HF}}^{(e)} = 2\pi\nu_{\text{HF}}^{(e)}$ places e_3 at zero and e_2 below it, and the leg drive is $\Omega_{ge} = \Omega\sqrt{3C_F^2(F_g, F_e)}$ from Eq. (1). The one-photon detuning Δ and the atomic velocity do *not* appear in H_0 ; they enter later as a single shift Δ_{eff} on the excited diagonal (Sec. 5), which is the trick that makes the Doppler average cheap.

Which field sits where (the branch). A subtlety unique to FWM: the pump and the weak fields can be assigned to the two legs in two inequivalent ways, corresponding to the two Raman resonances. The code labels them `branch =  $\mp 1$` . For the standard red-detuned seed (branch -1), the pump Ω_A drives the g_1 leg and the seed Ω_s drives the g_2 leg in H_0 ; for branch $+1$ the assignment swaps. These are *physically distinct mode pairs* and must be computed separately — never summed into one susceptibility matrix (see Sec. 7.3 and the README “Traps”).

4.2 The dissipator

The dissipator is a sum of GKSL terms, one per irreversible channel,

$$\mathcal{D}[\rho] = \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} L_k^\dagger L_k \rho - \frac{1}{2} \rho L_k^\dagger L_k \right). \quad (8)$$

Two kinds of channel appear (`atoms.AtomModel._build_lindblad`):

- **Spontaneous emission:** jump operators $L_{e \rightarrow g} = \sqrt{\Gamma_{e \rightarrow g}} |g\rangle\langle e|$ with the CG-branched rates of Eq. (2). The “recycling” term $L \rho L^\dagger$ refeeds the emitted population into the ground states; the anticommutator term damps the optical coherences at $\Gamma/2$ (the homogeneous optical linewidth).
- **Ground-state dephasing:** a phenomenological decay of the $g_1 \leftrightarrow g_2$ coherence, $\dot{\rho}_{g_1 g_2} \supset -\gamma_{gg} \rho_{g_1 g_2}$, with $\gamma_{gg}/2\pi = 100$ kHz (`constants.GAMMA_GG`). This is the rate that sets the width of the Raman/two-photon resonance and ultimately limits the squeezing bandwidth; it lumps together collisions, transit time, and laser phase noise.

Remark 1. The general *AtomModel* also supports polarization-grouped jump operators (Σ_q form) that transfer coherence during spontaneous emission (used by the magneto-optics schemes). The four-level FWM model only needs the population-refeeding form above.

4.3 Liouville space (vectorization)

Equation (6) is linear in ρ , so stacking the $n \times n$ matrix ρ into a length- n^2 vector turns the whole right-hand side into a single matrix — the Liouvillian $\mathcal{L}$. The code uses row-major vectorization, $\text{vec}(\rho)_{ni+j} = \rho_{ij}$, for which the commutator becomes (`core.comm_super`):

$$\text{vec}([H, \rho]) = (H \otimes \mathbb{K} - \mathbb{K} \otimes H^T) \text{vec}(\rho), \quad \mathcal{C}[H] \equiv -i(H \otimes \mathbb{K} - \mathbb{K} \otimes H^T). \quad (9)$$

The dissipator (8) vectorizes likewise; each term $L \rho L^\dagger$ becomes $L \otimes L^*$, and the anticommutators become $\frac{1}{2}(L^\dagger L \otimes \mathbb{K} + \mathbb{K} \otimes (L^\dagger L)^T)$ (`atoms._build_lindblad`). The full base Liouvillian is

$$\mathcal{L}_0 = \mathcal{C}[H_0] + \mathcal{D}, \quad (10)$$

(`core.build_liouvillian`). Finding the steady state is then the linear algebra problem $\mathcal{L}_0 \text{vec}(\rho) = 0$ subject to $\text{Tr} \rho = 1$; the trace condition replaces one redundant row of $\mathcal{L}_0$ (row 0, the $\rho_{g_1 g_1}$ equation) by $\sum_i \rho_{ii} = 1$. This “replace a row with the trace” construction recurs in every solver in `core.py`.

5 Thermal motion: the Doppler average

5.1 Velocity classes

At $T \sim 120^\circ\text{C}$ the atoms move fast enough that the Doppler shift kv ($\sigma_v = \sqrt{k_B T/m} \sim 170$ m/s gives $k\sigma_v/2\pi \sim 200$ MHz) dwarfs the natural linewidth. The vapour is therefore a Maxwell ensemble of velocity classes (`doppler.velocity_grid`):

$$v \in [-v_{\max}, v_{\max}], \quad v_{\max} = \lceil c_\sigma \sigma_v \rceil, \quad w(v) \propto \exp\left(-\frac{v^2}{2\sigma_v^2}\right), \quad \sum_v w(v) = 1, \quad (11)$$

with cutoff $c_\sigma = 3$ standard deviations and a step dv of a few m/s. Each class sees both lasers Doppler-shifted by kv (the pump and the near-degenerate seed/conjugate share essentially the same k , since they differ only by the GHz hyperfine scale $\ll$ the optical frequency).

5.2 The Δ_{eff} trick

The crucial observation (`doppler.py` header, README “Speed”) is that the *only* place the one-photon detuning Δ and the velocity v enter the Hamiltonian is through the excited-state diagonal. Define the effective one-photon detuning

$$\Delta_{\text{eff}} = \Delta - kv. \quad (12)$$

Then the velocity-dependent Liouvillian is a *rigid shift* of one template,

$$\mathcal{L}_0(\Delta_{\text{eff}}) = \mathcal{L}_0(0) - \Delta_{\text{eff}} S_v, \quad S_v = \mathcal{C} \left[\sum_{e \in \text{excited}} |e\rangle\langle e| \right], \quad (13)$$

where S_v is the commutator super-operator of the projector onto the excited manifold (`atoms._build_velocity_s`). Consequently:

- Temperature T enters *only* through the Maxwell weights $w(v)$.
- The detuning Δ enters *only* as an offset of the Δ_{eff} axis.
- The susceptibility table $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is therefore a function of $(\delta, \Delta_{\text{eff}})$ alone; it carries no explicit T or absolute Δ dependence.

The Δ_{eff} sampling axis is built (`doppler.build_Delta_eff_axis`) to span $\Delta - kv$ across a spectrum call’s velocity grid, so the later average is a cheap index interpolation rather than a re-solve. The dedicated (Δ, T) squeezing map builds a master Δ_{eff} axis from the exact union of all velocity samples needed by the scan and evaluates $\bar{\chi}(\delta, \Delta_{\text{eff}})$ once on that master grid; each map point then only gathers the relevant columns and applies the Maxwell weights.

6 The multimode steady state (truncated Floquet)

6.1 Why the steady state is not static

In ordinary absorption one weak field sits inside H_0 and the steady state is a single static ρ . FWM is different: the seed and the conjugate are *different frequencies*, separated by twice the ground hyperfine splitting, and they couple through the ground-state coherence $\rho_{g_1 g_2}$, which therefore *oscillates* at the beat frequency

$$\Omega_{\text{beat}} = \Omega_{\text{HF}} - \text{branch} \cdot \delta, \quad \Omega_{\text{HF}} = 2\pi\nu_{\text{HF}}. \quad (14)$$

The Hamiltonian is thus time periodic. We split it into the static part H_0 (pump + one weak field, Eq. (7)) and a *sideband* part that oscillates at Ω_{beat} (`sideband_hamiltonian`):

$$H(t) = H_0 + H_p e^{-i\Omega_{\text{beat}}t} + H_p^\dagger e^{+i\Omega_{\text{beat}}t}. \quad (15)$$

H_p contains the *other* weak field plus the pump on the complementary leg; its one-sided form ($|e\rangle\langle g|$ only, `_add_sideband_drive`) is the positive-frequency component of that drive.

6.2 Three-mode ansatz and truncation

Because the drive is weak, the response is dominated by the carrier and the first sidebands. Expand

$$\rho(t) = \rho_0 + \rho_{+1} e^{-i\Omega_{\text{beat}}t} + \rho_{-1} e^{+i\Omega_{\text{beat}}t} + \dots \quad (16)$$

and keep $|n| \leq 1$ (a three-mode truncated Floquet expansion). Substituting into the vectorized master equation and matching harmonics gives, with $C_p \equiv \mathcal{C}[H_p]$ and $C_m \equiv \mathcal{C}[H_p^\dagger]$ (`sideband.template`):

$$(\mathcal{L}_0 + i\Omega_{\text{beat}}) \rho_{+1} = -C_p \rho_0, \quad (17)$$

$$(\mathcal{L}_0 - i\Omega_{\text{beat}}) \rho_{-1} = -C_m \rho_0, \quad (18)$$

$$\mathcal{L}_0 \rho_0 = -(C_p \rho_{-1} + C_m \rho_{+1}). \quad (19)$$

Equation (19) is the carrier steady state *dressed* by the sidebands feeding back into it.

6.3 Feedback elimination

Solving Eqs. (17)–(18) for the sidebands and substituting into (19) eliminates $\rho_{\pm 1}$, leaving a single effective equation for the carrier (`core.floquet.solve`):

$$\boxed{\left[\mathcal{L}_0 - C_p (\mathcal{L}_0 - i\Omega_{\text{beat}})^{-1} C_m - C_m (\mathcal{L}_0 + i\Omega_{\text{beat}})^{-1} C_p \right] \rho_0 = 0.} \quad (20)$$

The two correction terms are exactly the code’s `minus_feedback` and `plus_feedback`; the bracket is `A_eff`. Trace-normalizing (row $0 \rightarrow \sum_i \rho_{0,ii} = 1$) and solving gives ρ_0 ; the upper sideband then follows directly,

$$\rho_{+1} = -(\mathcal{L}_0 + i\Omega_{\text{beat}})^{-1} C_p \rho_0, \quad (21)$$

reusing the factor already computed for (20) rather than solving a second system. Physically, ρ_0 holds the optical coherence at the seed frequency (the probe response) and ρ_{+1} holds the coherence at the conjugate frequency (the generated conjugate) — the two outputs we need.

Compiled grid solve. Equation (13), together with the affine δ dependence of the ground-state Hamiltonian, lets the FWM grid be assembled from constant matrices:

$$\mathcal{L}_0(\delta, \Delta_{\text{eff}}) = \mathcal{L}_0(0, 0) + \delta C_\delta - \Delta_{\text{eff}} S_v, \quad \Omega_{\text{beat}}(\delta) = \Omega_{\text{HF}} - \text{branch} \cdot \delta. \quad (22)$$

When Numba is available, `kernels.floquet_chi_grid` evaluates the full $(\delta, \Delta_{\text{eff}})$ grid in a fused parallel loop: assemble the local Liouvillian, factor the two sideband matrices, form A_{eff} , solve the trace-normalized carrier, recover ρ_{+1} , and contract directly with the CG-weighted readout vectors. The NumPy path follows the same equations with a per- δ loop over `core.floquet.solve`; `tests/test_kernels.py` pins the two paths to the same numerical result.

7 The susceptibility matrix

7.1 What $\bar{\chi}$ means

We want the linear response of the two optical polarizations (one per Raman leg) to the two weak fields. Define the dimensionless susceptibility matrix by

$$\begin{pmatrix} P_s \\ P_c \end{pmatrix} = \begin{pmatrix} \bar{\chi}_{ss} & \bar{\chi}_{sc} \\ \bar{\chi}_{cs} & \bar{\chi}_{cc} \end{pmatrix} \begin{pmatrix} \Omega_s \\ \Omega_c \end{pmatrix}, \quad (23)$$

where P_s (P_c) is the CG-weighted optical coherence read out on the seed (conjugate) leg, and Ω_s (Ω_c) is the seed (conjugate) Rabi frequency. The polarization readout is (`_polarization_coherence`):

$$P_{\text{leg } g} = \sum_{e \in \text{excited}} \sqrt{3C_F^2(F_g, F_e)} \rho_{eg}, \quad (24)$$

the same CG weighting as the drive, so absorption and emission are described by one consistent dipole. Because the response is linear in the *weak* fields (the pump is the strong field, already inside H_0 and the sidebands), $\bar{\chi}$ is *independent* of the magnitude of the reference field that probes it — the code simply divides the measured coherence by the reference Rabi Ω_{ref} .

7.2 Two solves per detuning

Each entry of the 2×2 matrix is a response of one leg to one field, so two independent Floquet solves at each two-photon detuning δ extract all four (`chi_matrix_table`):

- **Solve 1 — seed drive only** ($\Omega_c = 0$ in the sidebands, $\Omega_s = \Omega_{\text{ref}}$ in H_0). The carrier coherence on the seed leg gives $\bar{\chi}_{ss} = P_s/\Omega_{\text{ref}}$; the upper-sideband coherence on the conjugate leg gives $\bar{\chi}_{cs} = P_c/\Omega_{\text{ref}}$ — this is the seed $\rightarrow$ conjugate *generation*.
- **Solve 2 — conjugate seed only** ($\Omega_c = \Omega_{\text{ref}}$ in the sidebands, $\Omega_s = 0$). Symmetrically yields $\bar{\chi}_{sc}$ and $\bar{\chi}_{cc}$.

The off-diagonal $\bar{\chi}_{cs}, \bar{\chi}_{sc}$ are the parametric coupling that does not exist without the pump; they are the heart of the four-wave mixing. In the compiled path, these two physical solves correspond to two calls to `kernels.floquet_chi_grid`: one with the seed reference field and one with the conjugate reference field. Each call fills the whole $(\delta, \Delta_{\text{eff}})$ grid and returns the probe-leg and conjugate-leg coherences already contracted with the proper CG readout weights.

7.3 Detuning conventions and the two Raman branches

The probe is scanned in laboratory frequency, referenced to the $F = 2 \rightarrow F' = 3$ line. The two Raman resonances sit at (`branch_center_GHz`)

$$f_{\text{Raman}}^{(\pm)} = \Delta_{\text{GHz}} \pm \nu_{\text{HF}}, \quad (25)$$

and the two-photon detuning of a probe point is (`two_photon_detuning_from_probe_scan`)

$$\delta = 2\pi(f_{\text{probe}} - f_{\text{Raman}}^{(\text{branch})}). \quad (26)$$

The beat frequency (14) that drives the Floquet expansion uses this same δ . **The two branches are separate physical channels** (separate probe/conjugate mode pairs): the standard squeezing operating point is the red-detuned seed on the $(-)$ branch. Adding the two branches' $\bar{\chi}$ matrices into one propagation is wrong — it manufactures spurious high-gain features (the README documents an artefact near +70 MHz TPD). The wide “both branches” view computes branch -1 and $+1$ independently and overlays them (`full_spectrum`).

8 Doppler averaging of the susceptibility

The single-velocity table $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is averaged over the Maxwell distribution to give the macroscopic susceptibility seen by the beam. Using $\Delta_{\text{eff}} = \Delta - kv$ and the uniform Δ_{eff} axis, the average is a weighted linear interpolation (`doppler.doppler_average`):

$$\bar{\bar{\chi}}(\delta; \Delta, T) = \sum_v w(v) \bar{\chi}(\delta, \Delta - kv). \quad (27)$$

This is a *coherent average of the complex susceptibility*: amplitude and phase from each velocity class add before the field propagates, which is the correct order of operations for a dispersive

parametric medium (the alternative — averaging intensities — would discard the inhomogeneous phase that sets the gain). The interpolation indices and fractions are precomputed once (`interpolation_weights`) and applied to every row of the table (`apply_doppler_average`), so the average costs one matrix multiply. In the (Δ, T) squeezing-map script, the master Δ_{eff} axis is the exact union of all requested $\Delta - kv$ samples. The Doppler step therefore becomes a direct column gather followed by the same weighted sum, with no interpolation error and no additional Floquet solve.

9 Maxwell–Bloch propagation: susceptibility $\rightarrow$ gain

9.1 Coupled amplitude equations and phase conjugation

In a dilute medium the slowly varying seed and conjugate Rabi amplitudes obey coupled propagation equations along the cell (z direction). The combination that closes is $(\Omega_s, \Omega_c^*)^T$ — note the *complex conjugate* of the conjugate field, the signature of a phase-conjugating ($\chi^{(3)}$) process:

$$\frac{d}{dz} \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix} = M \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}, \quad M = \frac{i}{2} \begin{pmatrix} k_s \chi_{ss} & k_s \chi_{sc} \\ -k_c \chi_{cs}^* & -k_c \chi_{cc}^* \end{pmatrix}. \quad (28)$$

The diagonal terms are self-action (gain/absorption and dispersion on each beam); the off-diagonal terms are the parametric coupling that converts seed into conjugate and back. The conjugation and the minus signs in the second row are what make M generate *amplification* (a non-unitary, energy-non-conserving transfer) rather than mere rotation.

9.2 Seeded longitudinal phase mismatch

The seeded propagation matrix can carry a rotating-frame longitudinal mismatch term:

$$M_{\Delta k} = M + \frac{i \Delta k_z}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (29)$$

For the seeded double- Λ geometry the vacuum mismatch used in Balanced resolution is

$$\Delta k_z^{(0)}(\delta) = 2k_p - (k_s(\delta) + k_c(\delta)) \cos \theta, \quad \theta = 0.3^\circ, \quad (30)$$

where the reference angle is the Sim *et al.* pump–probe crossing angle, k_s is set by the scanned seed frequency, and energy conservation sets $k_c = 2k_p - k_s$ in angular-frequency units. This is cheap: it is one vector over the already-computed probe axis, and no OBE/Floquet solve is repeated.

Fine resolution adds two post-OBE refinements while keeping the expensive susceptibility table fixed:

1. A conservative refractive correction,

$$k_s \rightarrow n_s k_s, \quad k_c \rightarrow n_c k_c, \quad n_{s,c} \simeq 1 + \frac{1}{2} \text{Re } \chi_{ss,cc},$$

with the correction clipped to the dilute-vapour range so resonant dispersive excursions do not create artificial phase-matching runaways.

2. A segmented propagation profile. The cell is sliced into 16 segments; the diagonal self-action remains fixed, while the off-diagonal parametric coupling is smoothly attenuated by a scalar profile inferred from the median absorptive optical depth. This is a propagation-only approximation, not a new atomic solve.

The `phase_detail` option selects how much of this propagation physics is included. `PHASE_LEGACY` sets $\Delta k_z = 0$ for regression-preserving direct calculations. The app’s Fast and Balanced presets use Eq. (30); Fine uses the refractive and segmented refinements.

9.3 From dimensionless to physical susceptibility

The dimensionless $\bar{\chi}$ becomes a physical susceptibility through the medium coupling constant (`observables.gain_from_chi`):

$$\chi_{xy} = -\frac{2N|d_{\text{eff}}|^2}{\varepsilon_0\hbar}\bar{\chi}_{xy}, \quad |d_{\text{eff}}|^2 = \xi_{\text{LS}}|d_{D_1}|^2, \quad (31)$$

where N is the number density, d_{D_1} the D_1 reduced dipole, and ξ_{LS} is the residual *FWM coupling scale* (`LINE_STRENGTH_FACTOR`, app default ~ 0.05). The sign makes a passive transition absorptive ($\text{Im } \chi > 0$ on resonance). ξ_{LS} is the one remaining lumped calibration knob: the genuine hyperfine CG structure is already in $\bar{\chi}$, so ξ_{LS} stands in for residual not-yet-modelled physics (full propagation loss/noise, full Zeeman degeneracy, and other geometry details not captured by the seeded longitudinal Δk_z term). It rescales the *effective* $|d|^2$ in propagation only; it does *not* touch the OBE solve (where Γ and Ω are fixed by I_{sat}).

9.4 The transfer matrix

Because M is (per detuning point) a constant 2×2 matrix over the cell length L , the propagation is solved exactly by the matrix exponential

$$T = \exp(ML), \quad \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}_{z=L} = T \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}_{z=0}. \quad (32)$$

The 2×2 exponential is evaluated in closed form (`core.matrix_exp_2x2`; derivation in Appendix B) for the whole batch of detuning points at once. The power gains are

$$G_s = |T_{00}|^2 \quad (\text{seed transmission}), \quad G_c = |T_{10}|^2 \quad (\text{conjugate generated from a seed at port 0}). \quad (33)$$

With a seed in and vacuum in the conjugate port, G_s is the amplified seed power and G_c is the newly generated conjugate power. In the high-gain parametric regime the twin-beam relation $G_c \simeq G_s - 1$ holds (each created conjugate photon accompanies one added seed photon).

The exponential-density trap. Since $M \propto N$ (Eq. (31)) and $T = e^{ML}$, the gain grows *exponentially* with density, i.e. super-exponentially with temperature through $N(T)$. This is physically correct only while the pump is undepleted; pushed too far it over-amplifies, which the next section caps.

10 Pump depletion: the Manley–Rowe cap

The propagation above is linear in the *undepleted* pump, so at high density it can predict $G_s P_{\text{seed}} \gg P_{\text{pump}}$ — more output power than the pump can supply. Non-degenerate FWM is a Manley–Rowe process: two pump photons make one signal photon and one conjugate photon, so at full conversion the added signal power and the generated conjugate power each approach half the pump power. The code enforces this energy bound with a smooth homogeneous-saturation form (`observables.pump_depletion_saturation`):

$$G_s^{\text{sat}} = 1 + \frac{G_s - 1}{1 + (G_s - 1)P_{\text{seed}}/P_{\text{cap}}}, \quad G_c^{\text{sat}} = \frac{G_c}{1 + G_c P_{\text{seed}}/P_{\text{cap}}}, \quad P_{\text{cap}} = \frac{P_{\text{pump}}}{2}. \quad (34)$$

Three properties make this the right minimal fix:

- **Small-signal limit.** When $(G - 1)P_{\text{seed}} \ll P_{\text{cap}}$ the denominators are ≈ 1 and the validated gain is untouched.
- **Energy ceiling.** As $G \rightarrow \infty$, $(G_s^{\text{sat}} - 1)P_{\text{seed}} \rightarrow P_{\text{cap}}$ and $G_c^{\text{sat}}P_{\text{seed}} \rightarrow P_{\text{cap}}$ — exactly the half-pump bound.
- **Twin-beam relation preserved.** $(G_s - 1)$ and G_c saturate identically, so $G_c^{\text{sat}} \simeq G_s^{\text{sat}} - 1$ still holds and the squeezing formula stays consistent.

The pipeline keeps the pre-saturation small-signal peak for diagnostics and flags the operating point as “pump-depletion limited” when the small-signal gain exceeds the cap by more than 10 % (`compute_spectrum`, `_seeded_observables`).

11 Quantum observables of the twin beam

11.1 Intensity-difference squeezing

The twin beams are correlated, so the noise in $\hat{n}_s - \hat{n}_c$ can fall below the shot-noise level. For an ideal (lossless) two-mode parametric amplifier with gains G_s, G_c the intensity-difference noise, normalized to shot noise, is (`observables.intensity_difference_squeezing_dB`):

$$S_{\text{ideal}} = \frac{G_s + G_c - 2\sqrt{G_s G_c - 1}}{G_s + G_c}. \quad (35)$$

At unity gain $S_{\text{ideal}} = 1$ (shot noise); as the gain grows, $G_c \rightarrow G_s - 1$ and $S_{\text{ideal}} \rightarrow 0$ (perfect correlation). Symmetric detection efficiency η on both arms mixes in uncorrelated vacuum,

$$S(\eta) = \eta S_{\text{ideal}} + (1 - \eta), \quad S_{\text{dB}} = 10 \log_{10} S, \quad (36)$$

with $\eta = \text{QE}(1 - \text{loss})$. Equation (36) immediately gives the squeezing *floor*: since $S_{\text{ideal}} \geq 0$,

$$S_{\text{dB}} \geq 10 \log_{10}(1 - \eta), \quad (37)$$

a hard limit set purely by loss. (A 2-D $\xi(\Delta, T)$ scan confirms the model saturates exactly at this $10 \log_{10}(1 - \eta)$ plateau once the gain is large enough — the gain-saturation and the loss-floor together cap the achievable squeezing.) Negative S_{dB} means squeezed below shot noise; the Sim *et al.* reference point reaches about -8.33 dB in the focused scan.

11.2 Twin-beam photon statistics

In the same ideal-parametric spirit, the equal-time photon statistics follow from the mean pair number $n = G_s - 1$ (`observables.coincidence_stats`):

$$g_{ss}^{(2)} = g_{cc}^{(2)} = 2 \text{ (each arm thermal)}, \quad g_{sc}^{(2)}(0) = 2 + \frac{1}{n}, \quad R = \frac{[g_{sc}^{(2)}]^2}{g_{ss}^{(2)} g_{cc}^{(2)}} = \frac{(2 + 1/n)^2}{4}. \quad (38)$$

The Cauchy–Schwarz ratio $R > 1$ everywhere there is net gain is the nonclassical photon-pair signature; $g_{sc}^{(2)} \rightarrow 2$ at high flux and diverges at low flux. These are valid only where $G_s > 1$ (set to NaN otherwise) and, like the squeezing above, do not include quantum-Langevin propagation loss.

12 The seeded pipeline, end to end

Collecting the pieces, `compute_spectrum` (one one-photon detuning $\Delta = 2\pi D_{\text{GHz}} \cdot 10^9$) runs:

1. Convert powers to Rabi frequencies, Eq. (3); density $N(T)$.
2. Build the probe-frequency scan axis around the chosen Raman branch, and map each point to a two-photon detuning δ (Sec. 7.3).
3. Build the velocity grid and the Δ_{eff} axis (Sec. 5).
4. Build the $\bar{\chi}$ table on $(\delta, \Delta_{\text{eff}})$ (Secs. 6–7); with Numba available this is the fused `kernels.floquet_chi_grid` path, otherwise the NumPy `core.floquet_solve` reference path.
5. Doppler-average each of the four $\bar{\chi}$ entries, Eq. (27).
6. Maxwell–Bloch propagate $\rightarrow G_s, G_c$, Eqs. (28)–(33). In the app path the propagation matrix also carries the seeded phase-mismatch term, Eq. (29); Fine mode additionally uses the refractive and segmented propagation refinements.
7. Apply the Manley–Rowe cap, Eq. (34).
8. Squeezing $S_{\text{dB}}(\delta)$, Eq. (36).

The returned dictionary carries G_s, G_c, S_{dB} versus probe detuning, plus derived diagnostics ($N, \sigma_v, \Omega_{\text{pump}}, \Omega_{\text{seed}}$, the small-signal peak, the depletion cap, the active phase-detail mode, Δk_z , the vacuum Δk_z , and any segment profile diagnostics). The TPD slider in the app reads a single operating point off this curve by interpolation (`operating_point`) — instant, no re-solve — which is why the δ knob recomputes nothing.

Anchor. The direct `compute_spectrum` path remains regression-anchored. Around the *Sim et al.* reference setting ($\Delta = 0.9$ GHz, $T = 121$ °C, 600 mW pump, $8\mu\text{W}$ seed, 5.5 % loss), the focused two-photon-detuning scan returns $S_{\text{dB}} \approx -8.33$ dB with $G_s \approx 17.6$ at the local optimum. Regression tests hold the seeded gain/squeezing arrays to the frozen baseline, and the app presets select the appropriate phase-detail mode for interactive calculations.

13 Spontaneous FWM: the biphoton mode

The second mode (`GenericFWMSolver`) drops the seed entirely and estimates the *spontaneous* biphoton source: pairs generated from vacuum in a cascade or diamond level topology. It is a calibrated source model, not a Langevin propagation, but the physics it does contain is worth recording.

13.1 Topologies and fields

A topology is a set of `LevelSpec` levels and `FieldSpec` fields (pump, coupling, signal, idler), each with a wavelength, propagation direction, and phase sign. Three are provided:

- **⁸⁷Rb telecom cascade** $5S_{1/2} \rightarrow 5P_{3/2} \rightarrow 4D_{5/2}$ (780/1529 nm), calibrated to $\sim 38\,000$ cps/mW, $g^{(2)} \approx 44$, OD ≈ 112 , bandwidth ≈ 300 MHz.
- **¹³³Cs “BTW” cascade**, two channels (852–917 nm and 852–795 nm) that differ mainly in their temporal width.
- **Generic diamond** four-level template with user wavelengths (no validated anchor).

13.2 Energy and vector momentum conservation

Energy conservation across the four fields is a bookkeeping check (`energy_mismatch_hz`):

$$\Delta\nu = \nu_{\text{pump}} + \nu_{\text{coupling}} - \nu_{\text{signal}} - \nu_{\text{idler}} \approx 0. \quad (39)$$

Momentum conservation (phase matching) is evaluated as a vector mismatch (`phase_mismatch_vector`). The longitudinal component remains calibrated to the reference geometry,

$$\Delta k_z^{\text{rel}} = \Delta k_z^{\text{abs}} - \Delta k_{\text{ref}}, \quad \Delta k_z^{\text{abs}} = \sum_{\text{fields}} s_i k_i \cos \theta_i, \quad k_i = \frac{2\pi}{\lambda_i}, \quad (40)$$

with phase signs s_i and collection angles θ_i . The reference Δk_{ref} is applied only to the longitudinal component. The transverse component is absolute:

$$\Delta k_x = \sum_{\text{fields}} s_i k_i \sin \theta_i \sigma_i, \quad \sigma_i \in \{-1, 0, +1\}. \quad (41)$$

Pump and coupling fields use $\sigma_i = 0$ in the current biphoton collection model. Signal and idler use the selected collection side. Thus a longitudinally calibrated but transversely inconsistent setting, such as forcing the ^{87}Rb telecom source to $1.5^\circ/1.5^\circ$ on the same side, carries a large transverse mismatch and is suppressed by the sinc weight. The finite interaction length L turns this into a collection efficiency through the standard phase-matching sinc (`phase_matching_weight`):

$$\Delta k_{\text{vec}} = \sqrt{(\Delta k_z^{\text{rel}})^2 + \Delta k_x^2}, \quad \Phi_{\text{vec}} = \text{sinc}^2\left(\frac{\Delta k_{\text{vec}} L}{2}\right) = \left[\frac{\sin(\Delta k_{\text{vec}} L/2)}{\Delta k_{\text{vec}} L/2}\right]^2. \quad (42)$$

Scanning the idler angle traces out the one-dimensional acceptance lobe $\Phi_{\text{vec}}(\theta_{\text{idler}})$. In Fine phase-detail mode, the same vectorized geometry is evaluated on a signal–idler grid (`phase_mismatch_vector_grid`) producing a two-dimensional acceptance map $\Phi_{\text{vec}}(\theta_s, \theta_i)$ without touching the velocity-class source calculation.

13.3 Velocity-class coherent biphoton waveform

The temporal wavefunction of the pair is built as a *coherent sum over Maxwell velocity classes* (the hot-vapour analogue of the joint spectral amplitude). For a cascade, each class contributes a two-photon amplitude that is a product of the two intermediate-state resonances (`GenericFWMSolver.compute_biphoto`):

$$A(v) = w(v) \frac{\Omega_{\text{pump}} \Omega_{\text{coupling}}}{\left[\frac{\gamma_\ell}{2} + i(\Delta_{\text{pump}} - k_{\text{pump}} v)\right] \left[\frac{\gamma_u}{2} + i(\Delta_{2\gamma} - k_{\text{res}} v)\right]} \sqrt{\Phi}, \quad (43)$$

with γ_ℓ, γ_u the intermediate/upper linewidths, $\Delta_{2\gamma} = \Delta_{\text{pump}} + \Delta_{\text{coupling}}$ the two-photon detuning, and $k_{\text{res}} = k_{\text{pump}} + k_{\text{coupling}}$ the residual two-photon Doppler wavevector. The time-domain waveform is the coherent sum with the idler-arm propagation phase and the exponential emission envelope:

$$\psi(\tau) = \left[\sum_v A(v) e^{i k_{\text{idler}} v \tau} \right] e^{-\tau/\tau_{\text{decay}}}. \quad (44)$$

The first factor is a Doppler-dephasing sum (different velocity classes accumulate different phase $k v \tau$, so the pair correlation decays as the classes wash out); the second is the natural-lifetime envelope of the upper cascade state. The beating between the two cascade channels (e.g. the two Cs BTW channels) is exactly the velocity-class interference in Eq. (44) — which is why the two channels give different temporal widths.

13.4 Correlation readout and calibration

The detector model (`observables.biphoton_stats`) folds in finite filter transmission $\eta_f = \min(1, \text{BW}_{\text{filter}}/\text{BW}_{\text{source}})$, detector efficiencies $\eta_{s,i}$, dark counts, a coincidence window, and timing jitter (Gaussian smoothing of $|\psi|^2$). It then forms singles, true and accidental coincidences, and the normalized cross-correlation

$$g_{SI}^{(2)}(\tau) = 1 + \text{CAR} \cdot \frac{|\psi(\tau)|^2}{\max |\psi|^2}, \quad \text{CAR} = \frac{\text{coincidence}}{\text{accidental}}. \quad (45)$$

Heralding efficiencies and the Cauchy–Schwarz ratio $R = [g_{\text{peak}}^{(2)}]^2/4$ follow. Where a paper anchor exists (e.g. the ^{87}Rb $g^{(2)} \approx 44$), an explicit added-accidental term calibrates the peak to the reference value — making clear that the absolute noise floor is *calibrated*, not derived from first principles. A medium-complexity validation table checks the calibrated rate/OD/width anchors, energy bookkeeping, and phase matching.

14 Numerical architecture (brief)

The physics dictates the architecture, so a short note suffices:

- **One template, shifted.** $\mathcal{L}_0(\Delta_{\text{eff}}) = \mathcal{L}_0(0) - \Delta_{\text{eff}} S_v$ (Eq. (13)) lets every velocity share one Liouvillian and be assembled by cheap matrix shifts.
- **T - and Δ -independent table.** $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is computed on the needed grid; $T \rightarrow$ weights, $\Delta \rightarrow$ sampled columns or interpolation coordinates (Eq. (27)). The (Δ, T) map uses one exact-union master grid for all points.
- **Compiled OBE kernels.** `gabes/kernels.py` provides optional Numba kernels for the hot solve grids. FWM uses `floquet_chi_grid`; the absorption/Rydberg family uses the affine-scan kernel. The NumPy paths remain the reference fallback when Numba is unavailable.
- **Closed-form 2×2 .** The transfer matrix exponential is analytic (Appendix B), avoiding a per-point `expm`.
- **Thread hygiene.** The small dense systems are kept out of oversubscribed BLAS threading with `core.blas_single_thread`; compiled kernels parallelize at the grid level, and `full_spectrum` computes the two Raman branches independently.

15 Validation, scope, and what is *not* modelled

Validated. The direct seeded G_s/S_{dB} pipeline reproduces the Sim *et al.* ^{85}Rb operating point and is regression-locked bit-for-bit. The optional Numba kernels are tested against the NumPy reference paths, and the app-facing phase-detail modes add propagation physics without changing the atomic OBE definition. The genuine ^{85}Rb D_1 hyperfine CG factors (10, 35, 35, 28)/81 drive the coherent coupling, the polarization readout, and the spontaneous-emission branching.

Lumped / calibrated. A single residual coupling scale $\xi_{\text{LS}} \sim 0.05$ stands in for unmodelled propagation physics; the biphoton pair rates and the $g^{(2)}$ noise floor are reference-calibrated source estimates.

Not yet modelled (future physics).

- Self-consistent refractive phase matching. Fine mode reuses the computed susceptibility for a clipped, cheap refractive correction, but it does not re-solve the atom with propagation-modified fields.
- Full Gaussian spatial-mode overlap. Biphoton mode includes vector phase matching through Δk_z^{rel} and absolute Δk_x , but it still uses a scalar sinc acceptance rather than a full mode-overlap integral.
- Dynamical pump absorption/depletion along the cell. Fine mode uses a scalar segmented profile inferred from absorption, not a coupled pump-field Maxwell–Bloch solve.
- Explicit loss/noise: propagation loss with quantum-Langevin noise, seed excess noise, pump scattering, EOM residual multimode noise — currently all folded into the scalar detection efficiency η .
- Full Zeeman (m_F) substructure; the present model lumps the magnetic sublevels into the CG line strengths.

A Symbol glossary

symbol	meaning
Δ	one-photon (pump) detuning from $F = 2 \rightarrow F' = 3$
δ	two-photon (Raman) detuning
$\Delta_{\text{eff}} = \Delta - kv$	effective one-photon detuning of a velocity class
Ω	Rabi frequency (drive strength)
$\nu_{\text{HF}} = 3.0357 \text{ GHz}$	ground hyperfine splitting
$\Omega_{\text{beat}} = \Omega_{\text{HF}} \mp \delta$	ground-coherence beat (Floquet) frequency
$\Gamma/2\pi = 5.746 \text{ MHz}$	natural linewidth
$\gamma_{gg}/2\pi = 100 \text{ kHz}$	ground-coherence dephasing
C_F^2	hyperfine Clebsch–Gordan line strength
$\bar{\chi}_{xy}$	dimensionless susceptibility (leg x , field y)
χ_{xy}	physical susceptibility, Eq. (31)
N	atomic number density
T (matrix)	2×2 field transfer matrix $\exp(ML)$
G_s, G_c	seed / conjugate power gain
S, S_{dB}	intensity-difference squeezing (linear / dB)
$\eta = \text{QE}(1 - \text{loss})$	detection efficiency
ξ_{LS}	residual FWM coupling scale (<code>LINE_STRENGTH_FACTOR</code>)
Δk_z	seeded longitudinal phase mismatch
$\Delta k_z^{\text{abs}}, \Delta k_z^{\text{rel}}$	biphoton longitudinal absolute / calibrated relative mismatch
$\Delta k_x, \Delta k_{\text{vec}}$	biphoton transverse and vector phase mismatch
$\Phi_{\text{vec}} = \text{sinc}^2(\Delta k_{\text{vec}}L/2)$	biphoton vector phase-matching weight
$\psi(\tau)$	biphoton temporal waveform

B Closed-form 2×2 matrix exponential

Write $A = s\mathbb{K} + q$ with $s = \frac{1}{2} \text{Tr } A$ so that q is traceless. By the Cayley–Hamilton theorem a traceless 2×2 matrix satisfies $q^2 = -\det(q)\mathbb{K} = c^2\mathbb{K}$ with $c = \sqrt{-\det q}$. Hence

$$e^{qL} = \cosh(cL)\mathbb{K} + \frac{\sinh(cL)}{c}q, \quad e^{AL} = e^{sL} \left[\cosh(cL)\mathbb{K} + \frac{\sinh(cL)}{c}q \right]. \quad (46)$$

This is evaluated entrywise and batched over all detuning points in `core.matrix_exp_2x2`; the $c \rightarrow 0$ limit (where $\sinh(cL)/c \rightarrow L$) is handled explicitly to avoid the removable singularity. For the propagation matrix M of Eq. (28) this gives the transfer matrix $T = e^{ML}$ exactly, with no series truncation.

C File / function map

physics	function
Rabi from power	<code>rabi_freq</code>
density $N(T)$	<code>rb85_density</code> , <code>number_density</code>
CG line strengths	<code>CF2</code> , <code>line_strength</code> , <code>cf2</code>
level model + Lindblad	<code>AtomModel</code> , <code>_double_lambda_rb85</code>
commutator super-op	<code>comm_super</code>
Liouvillian	<code>build_liouvillian</code>
Hamiltonians	<code>static_hamiltonian_at_Deff_zero</code> , <code>sideband_hamiltonian</code>
Floquet steady state	<code>floquet_solve</code>
$\bar{\chi}$ matrix table	<code>chi_matrix_table</code> , <code>floquet_chi_grid</code>
velocity grid / Δ_{eff}	<code>velocity_grid</code> , <code>build_Delta_eff_axis</code>
Doppler average	<code>doppler_average</code>
Maxwell–Bloch gain / segmented propagation	<code>gain_from_chi</code>
2×2 matrix exp	<code>matrix_exp_2x2</code>
pump-depletion cap	<code>pump_depletion_saturation</code>
squeezing	<code>intensity_difference_squeezing_dB</code>
twin-beam statistics	<code>coincidence_stats</code>
full seeded pipeline	<code>compute_spectrum</code>
biphoton source	<code>GenericFWMSolver.compute_biphoton</code>
seeded Δk_z	<code>seeded_phase_mismatch_z</code>
phase matching	<code>phase_mismatch_vector</code> , <code>phase_mismatch_vector_grid</code> , <code>phase_matching_weight</code>
biphoton readout	<code>biphoton_stats</code>

Reference: G. Sim, H. Kim, H. S. Moon, *Sci. Rep.* **15**, 7727 (2025); Rb data from D. A. Steck, “Rubidium 85 D Line Data.”